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Sedra & Smith Microelectronic Circuits: Master Study Digest

High-Yield Summaries, Derivations & Formulas for GATE Analog/EDC

Reference: Sedra & Smith 8th Edition • **Target:** GATE Analog & EDC • **Value:** 18-22 Marks Direct

1. MOS Field-Effect Transistor (MOSFET) Small-Signal Model

- **Transconductance Formulations:** Three mathematically equivalent representations of g_m in saturation:
  1. g_m = mu_n C_ox (W/L) V_ov
  2. g_m = sqrt(2 mu_n C_ox (W/L) I_D)
  3. g_m = 2 I_D / V_ov

where V_ov = V_GS - V_th is the overdrive voltage.

- **Output Resistance:** r_o = |V_A| / I_D = 1 / (lambda * I_D). Intrinsic MOSFET gain A_0 = g_m * r_o = 2 |V_A| / V_ov.

- **Body Effect:** V_th = V_th0 + gamma * [sqrt(2 phi_F + V_SB) - sqrt(2 phi_F)]. Increases threshold voltage when source is at a higher potential than body.

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Transconductance	**g_m = 2 I_D / V_{ov} = sqrt(2 k_n I_D)**	I_D = Bias, V_ov = Overdrive	Primary voltage-to-current conversion factor
Intrinsic Gain	**A_0 = g_m * r_o = 2 V_A / V_{ov}**	V_A = Early Voltage	Maximum theoretical voltage gain achievable by a single transistor
T-Model Resistance	**r_e = 1 / g_m**	Source Resistance	Resistance seen looking directly into the source terminal

2. Differential Pairs & Common-Mode Rejection Ratio (CMRR)

- **MOS Differential Pair:** Differential gain A_d = - g_m (R_D || r_o). Common-mode gain with tail current source resistance R_{SS}: A_{cm} = - (g_m R_D) / (1 + 2 g_m R_{SS}) ≈ - R_D / (2 R_{SS}).

- **CMRR Definition:** CMRR = |A_d / A_{cm}| ≈ 2 g_m R_{SS}. Expressed in decibels: CMRR_{dB} = 20 log10 |A_d / A_{cm}|. To maximize CMRR, tail resistance R_{SS} must be maximized (e.g. using a cascode current mirror).

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Differential Gain	**A_d = - g_m * R_D**	R_D = Drain Load	Gain applied to pure differential input V_{id} = V_1 - V_2
Common Mode Gain	**A_{cm} ≈ - R_D / (2 R_{SS})**	R_SS = Tail Current Impedance	Unwanted amplification of identical noise on both inputs
CMRR (Linear / dB)	**CMRR = 2 g_m R_{SS} | 20 log_{10}(CMRR)**	g_m = Transconductance	Measure of amplifier ability to reject common-mode noise